

Numerical Method: Bisection $f(x) = 2(x - x_0) + 2(x^2 + 5 - y_0)(2x)$
(0')

At $\{x_0, y_0\} = \{0, 0\}$:

$$f(x) = 2x + 2(x^2 + 5)(2x)$$

At $x_1 = 1$, $f(x_1) = 2(1) + 2(1+5)(2) = 26$

$x_2 = -1$, $f(x_2) = 2(-1) + 2(-1+5)(-2) = -18$

Since $f(x_1)$ and $f(x_2)$ change signs, choose this interval.

$$x_3 = \frac{x_1 + x_2}{2} = 0$$

$$f(0) = 2(0) + 2(0^2 + 5)(2(0)) = 0 \quad \checkmark \quad x = 0$$

$$y = 0^2 + 5 = 5$$

$$d = \sqrt{0^2 + 5^2} = 5$$

At $\{x_0, y_0\} = \{-4, 0\}$:

$$f(x) = 2(x + 4) + 2(x^2 + 5)(2x)$$

At $x_1 = 1$, $f(x_1) = 2(1+4) + 2(1+5)(2) = 34$

$x_2 = -1$, $f(x_2) = 2(-1+4) + 2((-1)^2 + 5)(-2) = -18$

Since $f(x_1)$ and $f(x_2)$ change signs, choose this interval.

$$x_3 = \frac{x_1 + x_2}{2} = 0$$

$$f(0) = 2(0+4) + 2(0+5)(2(0)) = 8 > 0, \text{ test } x_2 \rightarrow x_3$$

$$x_4 = \frac{x_2 + x_3}{2} = \frac{-1 + 0}{2} = -\frac{1}{2}$$

$$f(-\frac{1}{2}) = 2(-\frac{1}{2} + 4) + 2((-\frac{1}{2})^2 + 5)(2(-\frac{1}{2})) = -\frac{7}{2} < 0$$

$$x = -0.5 \quad y = (-0.5)^2 + 5 = 5.25$$

$$d = \sqrt{(-0.5+4)^2 + 5.25^2} = 6.30 //$$

At $\{x_0, y_0\} = \{-8, 0\}$:

$$f(x) = 2(x + 8) + 2(x^2 + 5)(2x)$$

At $x_1 = 1$, $f(x_1) = 2(1+8) + 2(1+5)(2) = 32$

$x_2 = -1$, $f(x_2) = 2(-1+8) + 2(-1+5)(-2) = -2$

Since $f(x_1)$ and $f(x_2)$ change signs, choose this interval.

$$x_3 = \frac{x_1 + x_2}{2} = 0$$

$$f(0) = 2(0+8) + 2(0+5)(2(0)) = 16 > 0, \text{ test } x_2 \rightarrow x_3$$

Iteration 1

Iteration 2.
stop here.

Iteration 1

$$x_4 = \frac{x_2 + x_3}{2} = \frac{-1 + 0}{2} = -\frac{1}{2}$$

$$f(-\frac{1}{2}) = 2(-\frac{1}{2} + 8) + 2(-\frac{1}{2} + 5)(2(-\frac{1}{2})) = 6 > 0$$

Iteration 2. stop here

$$x = -0.5$$

$$y = (-0.5)^2 + 5 = 5.25$$

$$d = \sqrt{(-0.5 + 8)^2 + 5.25^2} = 9.155 //$$

At $x_0, y_0 = (2, 0)$:

$$f(x) = 2(x - 2) + 2(x^2 + 5)(2x)$$

At $x_1 = 1$, $f(x_1) = 2(1 - 2) + 2(1 + 5)(2) = 22$

$x_2 = -1$, $f(x_2) = 2(-1 - 2) + 2((-1)^2 + 5)(-2) = 18$

Since $f(x_1)$ and $f(x_2)$ change signs, choose this interval.

Iteration 1

$$x_3 = \frac{x_1 + x_2}{2} = 0$$

$$f(0) = 2(0 - 2) + 2(0 + 5)(2(0)) = -4 < 0, \text{ test } x_1 \rightarrow x_3$$

$$x_4 = \frac{x_1 + x_3}{2} = \frac{1 + 0}{2} = \frac{1}{2}$$

Iteration 2. stop here.

$$f(\frac{1}{2}) = 2(\frac{1}{2} - 2) + 2((\frac{1}{2})^2 + 5)(2(\frac{1}{2})) = \frac{15}{2} > 0$$

$x = -0.5$ $y = (-0.5)^2 + 5 = 5.25$

$$d = \sqrt{(-0.5 - 2)^2 + 5.25^2} = 5.815$$

At $x_0, y_0 = (6, 0)$:

$$f(x) = 2(x - 6) + 2(x^2 + 5)(2x)$$

At $x_1 = 1$, $f(x_1) = 2(1 - 6) + 2(1 + 5)(2) = 34$

$x_2 = -1$, $f(x_2) = 2(-1 - 6) + 2((-1)^2 + 5)(-2) = -38$

Since $f(x_1)$ and $f(x_2)$ change signs, choose this interval.

Iteration 1

$$x_3 = \frac{x_1 + x_2}{2} = 0$$

$$f(0) = 2(0 - 6) + 2(0 + 5)(2(0)) = -12 < 0, \text{ test } x_1 \rightarrow x_3$$

$$x_4 = \frac{x_2 + x_3}{2} = \frac{-1 + 0}{2} = -\frac{1}{2}$$

Iteration 2. stop here.

$$f(-\frac{1}{2}) = 2(-\frac{1}{2} - 6) + 2((-\frac{1}{2})^2 + 5)(2(-\frac{1}{2})) = -\frac{47}{2} < 0$$

$x = -0.5$ $y = (-0.5)^2 + 5 = 5.25$

$$d = \sqrt{(-0.5)^2 + 5.25^2} = 5.274$$

$$d = \sqrt{(-0.5 - 6)^2 + 5.25^2} = 8.355 //$$